

Behavioral Calibration for Mathematical Reasoning under Low-Data Reinforcement Learning

Controlled Experiments and Token-Level Evidence for 1-shot GRPO

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Central Goal

Explain whether 1-shot GRPO produces a real held-out gain, where that gain comes from, and what mechanism best explains it.

- 1 **Does it help?** Does 1-shot GRPO improve MATH500 accuracy beyond the initial model?
- 2 **Where does the gain come from?** Is it tied to reward semantics, data budget, group-relative signal, or model family?
- 3 **What is the mechanism?** Does training create new knowledge, or calibrate how existing reasoning behavior is expressed?

Takeaway

The evidence supports a calibration account: 1-shot GRPO helps the Base model when the reward is meaningful, mainly by stabilizing the generation distribution.

Controlled Comparisons

- Model family: Base vs. Instruct
- Data budget: 1-shot vs. full-data
- Group signal: $n=8$ vs. $n=1$
- Regularization: standard vs. weak-reg.
- Reward semantics: clean vs. random reward

Evidence Layers

- Score curves answer whether the gain exists.
- Matched controls identify the source of the gain.
- Token metrics test the calibration mechanism.
- Full-token traces show the mechanism inside a response.

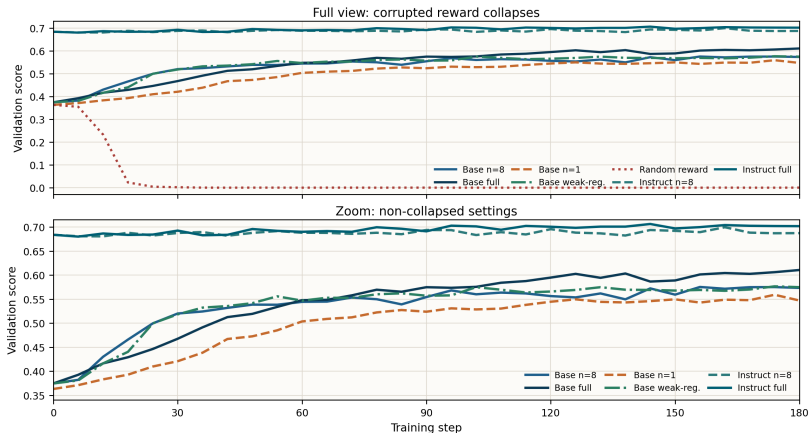
Controls for the Three Questions

Setting	Question Tested	Interpretation
Base n=8	Does 1-shot help?	Main low-data effect on Base model
Base full-data	Is 1-shot enough?	Matched data-rich reference
Base n=1	Source of the signal?	Role of grouped rollouts
Base weak-reg.	Only regularization?	Sensitivity to entropy/KL pressure
Base random-reward	Does reward meaning matter?	Reward semantics destroyed
Instruct n=8	Does model family matter?	Effect after instruction tuning
Instruct full-data	Does more data help Instruct?	Instruct data-rich reference

Takeaway

Each control isolates one possible source of the 1-shot GRPO gain.

Question 1: Does 1-Shot GRPO Help?



Base 1-shot GRPO improves strongly; corrupted rewards collapse, so the effect is not generic RL.

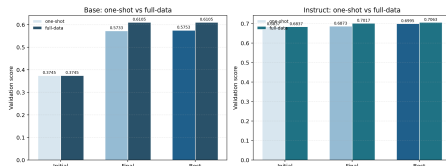
Answer 1: The Gain Is Real but Conditional

Setting	Best Acc.	Final Acc.	Role
Base n=8	0.603	0.583	Main low-data result
Base full-data	0.611	0.611	Full-data Base reference
Base n=1	0.559	0.547	Group-size ablation
Base weak-reg.	0.577	0.575	Regularization ablation
Base random-reward	0.363	0.000	Strict negative control
Instruct n=8	0.700	0.687	1-shot Instruct result
Instruct full-data	0.706	0.702	Full-data Instruct reference

Takeaway

1-shot GRPO gives a substantial Base-model gain, but the gain depends on meaningful rewards and does not remove the value of full data.

Question 2: How Much Comes From Data Budget?



Interpretation

- Base final gap: $+0.0373$ for full-data over one-shot.
- Instruct final gap: $+0.0145$.
- One-shot captures a clean low-data gain, but full-data still improves the endpoint.

Question 3: What Changes in the Output Distribution?

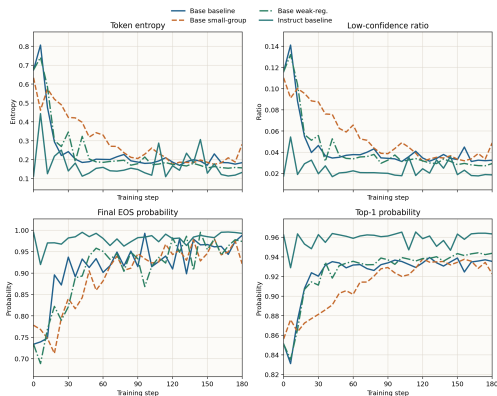
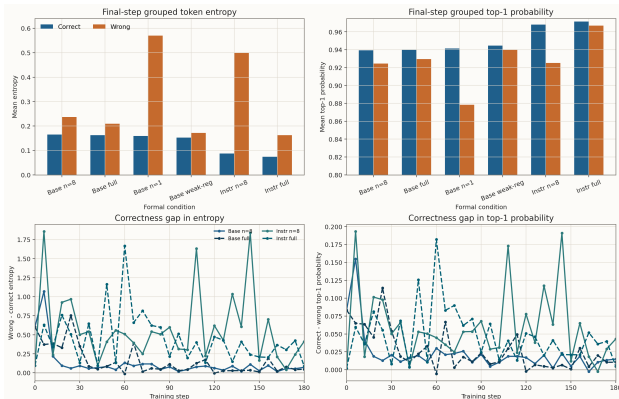


Figure 1: Length, entropy, top-1 probability, EOS probability, and tail confidence.

Successful settings move toward shorter, lower-entropy, higher-confidence generation.

Mechanism Signal: Correct Responses Are Better Calibrated



Correct responses are systematically lower-entropy and higher-confidence than wrong responses across the non-collapse settings. This links score gains to distributional stabilization, not only endpoint accuracy.

Protocol

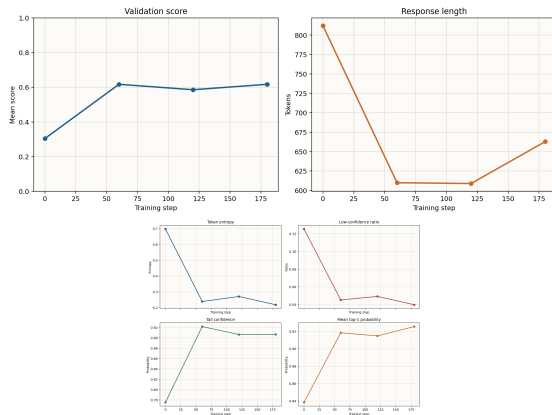
Base n=8 1-shot checkpoints at steps 0, 60, 120, 180; 32 held-out problems; 4 sampled responses per problem; every generated token is scored.

Step	Score	Len.	Ent.	Low-conf.	Top-1
0	0.305	812.1	0.698	0.125	0.839
60	0.617	609.9	0.239	0.045	0.918
120	0.586	608.9	0.271	0.049	0.915
180	0.617	662.9	0.219	0.040	0.926

Takeaway

The main behavioral transition appears early: higher score, shorter responses, lower entropy, and fewer low-confidence tokens.

Mechanism Over Checkpoints



The main transition occurs by step 60: score rises while length, entropy, and low-confidence mass fall sharply. This looks like rapid calibration, not slow knowledge acquisition.

Token entropy over generated reasoning: MATH500 #18, sample 0
step 180 | score=1.000 | mean entropy=0.259 | length=485

Given: $\overline{BC}$ is parallel to the segment through A . $AB = BC$. We need to find the number of degrees represented by x . Since $\overline{BC}$ is parallel to the segment through A , we can use the properties of parallel lines and transversals. The angle $\angle ABC$ and the given angle 124° are corresponding angles, meaning they are equal. Let's deduce the measure of $\angle ABC$: 1. Since $\overline{BC}$ is parallel to the segment through A , and $\angle ABC$ and 124° are corresponding angles, we have: $\angle ABC = 124^\circ$ 2. Since $AB = BC$, triangle ABC is isosceles, and thus $\angle BAC = \angle BCA$. 3. The sum of the interior angles in a triangle is 180° . Hence: $\angle BAC + \angle BCA + \angle ABC = 180^\circ$ 4. Given that $\angle BAC = \angle BCA$, let: $\angle BAC = \angle BCA = x$ So, we have: $x + x + 124^\circ = 180^\circ$ which simplifies to: $2x + 124^\circ = 180^\circ$ Solving for x : $2x = 180^\circ - 124^\circ$ $2x = 56^\circ$ $x = \frac{56^\circ}{2}$ $x = 28^\circ$ Therefore, the number of degrees represented by x is 28° . The final answer is: $\boxed{28}$

Wrong trace: high entropy around the phase error

step 180 | score=0.000 | GT= π | output= $\pi/2$ | mean H=0.313 | len=296

The function given is $y = a \sin(bx + c) + d$. From the graph, we can deduce some information about the parameters: 1. The amplitude of the function is 2, so $a = 2$. 2. The graph seems to complete one full wave every $\frac{2\pi}{3}$, so the period $T = \frac{2\pi}{3}$. The period of a sine function is given by $T = \frac{2\pi}{b}$, so $b = 3$. 3. The graph reaches its maximum value at $x = 0$, which is $d + a = 3$. Since $a = 2$, we get $d + 2 = 3$, so $d = 1$. 4. The graph reaches its maximum value of 3 at $x = 0$, which means $\sin(c) = 1$. The sine function reaches its maximum value of 1 at $\frac{\pi}{2} + 2k\pi$ for any integer k . So, $c = \frac{\pi}{2}$. The smallest possible value of c is $\frac{\pi}{2}$. The final answer is: $\boxed{\frac{\pi}{2}}$

Failure point: It treats $x=0$ as a maximum. But $y(0)=2 \sin(\pi)+1=1$, so $x=0$ is the midline; the smallest c is π .

low entropy high entropy

High-H tokens: graph reaches / seems / phase

Why the Gain Looks Like Calibration

- The gain is not produced by longer exploration: step 60 is much shorter than step 0.
- Step 60 and step 180 tie in score but differ behaviorally, so no single token metric explains everything.
- At the final checkpoint, correct responses remain lower-entropy, while wrong traces can reveal high-entropy mistaken premises.
- EOS pressure drops after training and only partly rebounds at step 180, so the gain is not just early stopping.

Takeaway

The mechanism is a coordinated shift in generation behavior: confidence, entropy, length, and stopping behavior move together.

Source of the Gain: Reward Semantics Matter



Reading

- Successful runs reduce uncertainty without matching the corrupted-reward collapse.
- Random-reward training moves into an unstable distributional regime.
- Weak regularization remains competitive, so the story is not simply stronger entropy suppression.

1. Does 1-shot GRPO have a benefit?

Yes for the Base model: Base $n=8$ reaches 0.603 best accuracy and 0.583 final accuracy.

2. Where does the benefit come from?

It depends on meaningful reward semantics and grouped rollout comparison; full-data still improves the endpoint, so one-shot is useful but not a complete substitute.

3. What is the mechanism?

The evidence points to behavioral calibration: shorter responses, lower entropy, higher top-1 probability, fewer low-confidence tokens, and trace-level entropy around reasoning choices.

Thank You